POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA
INDUSTRIALE E DELL'INFORMAZIONE

Ingegneria Fisica - Engineering Physics

Info-event on LM

10 June, 2025

www.ccs-fisica.polimi.it

New Student representatives (since 28/5/2025)

CONSIGLIO DEL CORSO DI STUDIO IN INGEGNERIA FISICA

1. Lorenzo De Lotto
2. Gabriele Facchini
3. Elena Foppoli
4. Stefano Galli
5. Jessica Lin
6. Giovanna London
7. Andrea Palucci
8. Sofia Ravetta
9. Luca Riva

Summary

- Update on Tracks/PSPA for a.y. 2025/26 (Prof. Ghiringhelli)
- *Management of Innovation Projects* (Prof. Cagliano, DIG)
- “Luxottica”: Interdisciplinary Program on Smart Wearable Technologies (Prof. Della Valle)
- Admissions and Study Plans (Prof. Zavelani Rossi, Prof. Dallera)
- International exchanges (Prof. Bottegoni)

- Q & A

General structure of the study plan

Year	MANDATORY COURSES	At choice fom a list of courses	CFU
1		Mathematics	5
1	SEMICONDUCTOR PHOTONICS		10
	PHOTONICS AND RADIOMETRY		
1	SOLID STATE PHYSICS		10
1		Computer science	10
1	ELECTRONICS		10
1	Physics or Engineering, depends on track		10
1		Choice within groups ENG, PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	10
2		Physics, chosen in a list specific to each track	10-20
2	Math, Engineering, Physics, depends on track		0-10
2		Choice within groups ENG, PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	10-20
2	THESIS WORK		15

65 CFU

40 CFU

15 CFU

Track = PSPA = *Piano degli Studi Preventivamente Approvato*

Structure of the study program

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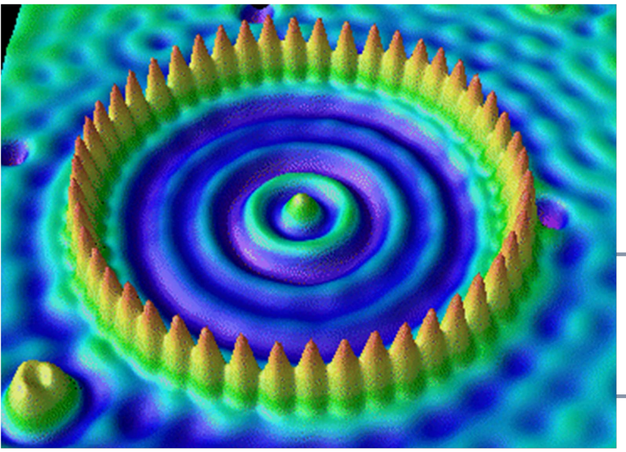
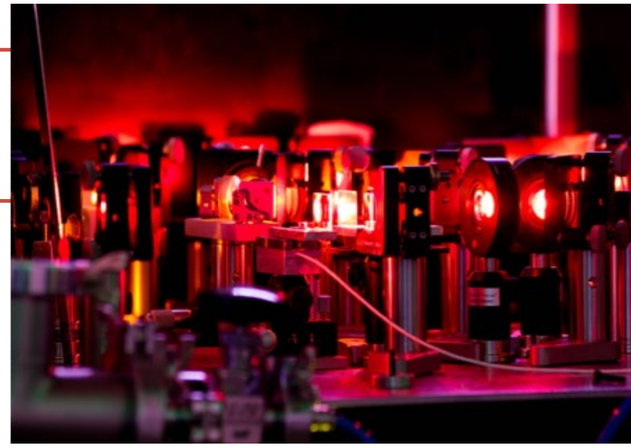
4 tracks (PSPA) are suggested addressing the main areas of modern physics with high technological/industrial impact

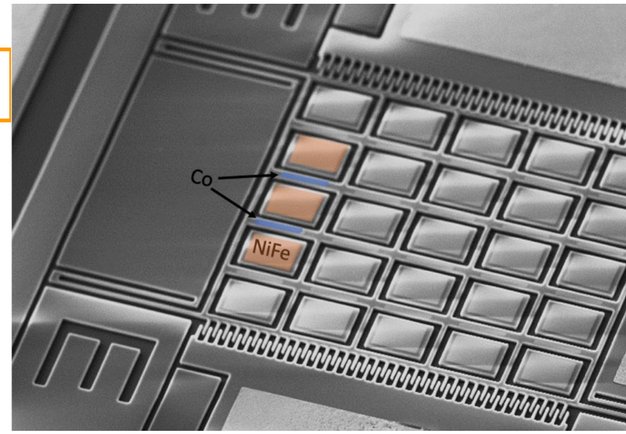


F2A PHOTONIC AND QUANTUM TECHNOLOGIES



QUANTUM MATERIALS AND NANOPHYSICS **F2B**

F2C MICRO AND NANO SYSTEMS



MATHEMATICS AND PHYSICS FOR QUANTUM ENGINEERING **F2D**

1st year

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F2A

Photonic and Quantum Technologies

Exam n.	Course Title	Semester	CFU	
1	NUMERICAL METHODS FOR ENGINEERING	1	5	5
	APPLIED STATISTICS	2	5	
	MECCANICA ANALITICA	1	5	
2	SEMICONDUCTOR PHOTONICS	1	5	10
	PHOTONICS AND RADIOMETRY	1	5	
3	SOLID STATE PHYSICS	1	10	
4	DIGITAL SYSTEMS DESIGN	1	10	10
	SYSTEMS THEORY (NONLINEAR DYNAMICS)	1	5	
	DATA MINING	1	5	
	ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING	1	5	
	ONLINE LEARNING APPLICATIONS	2	5	
	STREAMING DATA ANALYTICS	1	5	
5	ELECTRONICS	2	10	
6	INTEGRATED AND META PHOTONICS [I.C.]	2	10	
7	Courses to be chosen from Groups ENG, PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	--	10	

2nd year

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F2A

Photonic and Quantum Technologies

Exam n.	Course Title	Sem.	CFU	
1-3	PHYSICS AND APPLICATIONS OF ULTRAFAST PROCESSES [I.C.]	1	10	20
	PHYSICS OF ULTRAFAST PROCESSES	1	5	
	APPLICATIONS OF ULTRAFAST PROCESSES	1	5	
	QUANTUM OPTICS AND QUANTUM TECHNOLOGIES [I.C.]	2	10	
	QUANTUM OPTICS AND INFORMATION	2	5	
	ATOM OPTICS AND QUANTUM TECHNOLOGIES	2	5	
	BIOPHOTONICS [I.C.]	1	10	
	OPTICAL MICROSCOPY	1	5	
	BIOMEDICAL OPTICS	1	5	
	METHODS FOR MULTIMODAL FUNCTIONAL EVALUATION OF THE CENTRAL NERVOUS SYSTEM (o)	2	5	
	APPLICATIONS OF OPTICAL MICROSCOPY IN BIOMEDICINE (o)	2	5	
4-6	Courses to be chosen from Groups ENG, PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	--	20	
	THESIS WORK	1,2	15	

Note: (o) Limited enrollment course

1st year

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Quantum Materials and Nanophysics

F2B

Exam n.	Course Title	Semester	CFU	
1	NUMERICAL METHODS FOR ENGINEERING	1	5	5
	APPLIED STATISTICS	2	5	
	MECCANICA ANALITICA	1	5	
2	SEMICONDUCTOR PHOTONICS	1	5	10
	PHOTONICS AND RADIOMETRY	1	5	
3	SOLID STATE PHYSICS	1	10	
4	DIGITAL SYSTEMS DESIGN	1	10	10
	SYSTEMS THEORY (NONLINEAR DYNAMICS)	1	5	
	DATA MINING	1	5	
	ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING	1	5	
	ONLINE LEARNING APPLICATIONS	2	5	
	STREAMING DATA ANALYTICS	1	5	
5	ELECTRONICS	2	10	
6	QUANTUM PHENOMENA IN LOW DIMENSIONAL SYSTEMS	2	5	10
	STATISTICAL PHYSICS OF COMPLEX SYSTEMS	2	5	
7	Courses to be chosen from Groups ENG, PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	--	10	

2nd year

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F2B

Quantum Materials and Nanophysics

Exam n.	Course Title	Sem.	CFU	
1-3	MAGNETISM, SUPERCONDUCTIVITY AND SPINTRONICS [I.C.]	1	10	20
	NANOMAGNETISM AND SPINTRONICS	1	5	
	MAGNETISM AND SUPERCONDUCTIVITY	1	5	
	SEMICONDUCTOR NANOSTRUCTURES	1	5	
	QUANTUM DEVICES AND NANOELECTRONICS [I.C.]	1	10	
	NANOELECTRONICS OF GRAPHENE AND RELATED 2D MATERIALS	1	5	
	SEMICONDUCTOR QBITS	1	5	
	ELECTRON AND SCANNING PROBE MICROSCOPY	2	5	
4-6	Courses to be chosen from Groups ENG, PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	--	20	
	THESIS WORK	1,2	15	

1st year

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F2C

Micro and Nano Systems

Exam n.	Course Title	Semester	CFU	
1	NUMERICAL METHODS FOR ENGINEERING	1	5	
2	SEMICONDUCTOR PHOTONICS	1	5	10
	PHOTONICS AND RADIOMETRY	1	5	
3	SOLID STATE PHYSICS	1	10	
4 4	DIGITAL SYSTEMS DESIGN	1	10	10
	SYSTEMS THEORY (NONLINEAR DYNAMICS)	1	5	
	DATA MINING	1	5	
	ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING	1	5	
	ONLINE LEARNING APPLICATIONS	2	5	
	STREAMING DATA ANALYTICS	1	5	
	MACHINE DESIGN	2	5	
5	ELECTRONICS	2	10	
6	STRUCTURAL MECHANICS	2	5	10
	SEMICONDUCTOR DEVICE MANUFACTURING	1	5	
	PRODUCTION PLANNING AND CONTROL FOR MICRO/NANO SYS.	2	5	
7	Courses to be chosen from Group ENG	--	10	

2nd year

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F2C

Micro and Nano Systems

Exam n.	Course Title	Sem.	CFU	
1-2	MAGNETISM AND SUPERCONDUCTIVITY	1	5	10
	NANOMAGNETISM AND SPINTRONICS	1	5	
	ELECTRON AND SCANNING PROBE MICROSCOPY	1	5	
	PHYSICS OF SEMICONDUCTORS	1	5	
	SEMICONDUCTOR NANOSTRUCTURES	1	5	
	NANOELECTRONICS OF GRAPHENE AND RELATED 2D MATERIALS	1	5	
	SEMICONDUCTOR QBITS	1	5	
	INTEGRATED PHOTONICS	1	5	
	METAPHOTONICS	1	5	
	ANALOG AND DIGITAL HOLOGRAPHY	1	5	
	PHYSICS AND TECHNOLOGY OF OPTICAL MATERIALS	1	5	
3-5	Courses to be chosen from Groups MOD, M&D, MAN, SOFT, INT	-	20	
6	ADVANCED MICRO AND NANOFABRICATION TECHNOLOGIES	1	5	5
7	MULTI-DISCIPLINARY DESIGN LABORATORY	2	5	5
	THESIS WORK	1,2	15	

1st year

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Mathematics and Physics for Quantum Engineering

F2D

Exam n.	Course Title	Semester	CFU	
1	MECCANICA ANALITICA [(a)]	1	5	5
2	SEMICONDUCTOR PHOTONICS	1	5	10
	PHOTONICS AND RADIOMETRY	1	5	
3	SOLID STATE PHYSICS	1	10	
4	DIGITAL SYSTEMS DESIGN	1	10	10
	SYSTEMS THEORY (NONLINEAR DYNAMICS)	1	5	
	DATA MINING	1	5	
	ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING	1	5	
	ONLINE LEARNING APPLICATIONS	2	5	
	STREAMING DATA ANALYTICS	1	5	
5	ELECTRONICS	2	10	
6	GROUP THEORY	2	5	
7	OPERATOR THEORY	1	5	
8	Courses to be chosen from Groups PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	--	10	

[(a)] To be replaced with a different MAT exam for those comeing from Mathematical Engineering

2nd year

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Mathematics and Physics for Quantum Engineering

F2D

Exam n.	Course Title	Semester	CFU	
1	MATHEMATICS OF QUANTUM MECHANICS	1	10	
2	MAGNETISM AND SUPERCONDUCTIVITY	1	5	5
	SEMICONDUCTOR QBITS	1	5	
	PHYSICS OF ULTRAFAST PROCESSES	1	5	
	SYMMETRY APPLICATIONS IN CLASSICAL & QUANTUM PHYSICS	2	5	
	ATOM OPTICS AND QUANTUM TECHNOLOGIES	2	5	
	META PHOTONICS	2	5	
	STATISTICAL PHYSICS OF COMPLEX SYSTEMS	2	5	
3	QUANTUM CIRCUITS AND DEVICES	1	10	10
	QUANTUM COMPUTING	1	5	
	HIGH INTENSITY LASERS FOR NUCLEAR AND PHYSICAL APPL. II	2	5	
	PHYSICAL PROPERTIES OF MOLECULAR MATERIALS	1	5	
	MOLECULAR MODELING OF MATERIALS	2	5	
4	QUANTUM OPTICS AND INFORMATION	2	5	
5	Courses to be chosen from Groups PHYS-1, PHYS-2, MATH, MAN, SOFT, INT	--	10	
	THESIS WORK	1,2	15	

Group ENG (Engineering)

SSD	Denominazione Insegnamento	Sem	CFU
ING-INF/04	CONTROL OF INDUSTRIAL ROBOTS	1	5.0
ING-INF/05	DIGITAL SYSTEMS DESIGN	1	10.0
ING-INF/01	ELECTRON DEVICES	1	10.0
ING-IND/22	FUNCTIONAL MATERIALS	1	5.0
ING-INF/01	MEMS AND MICROSENSORS	1	10.0
ING-IND/22	MICROSTRUCTURAL CHARACTERISATION OF MATERIALS	1	5.0
ING-INF/04 ING-INF/07	OPTICAL MEASUREMENTS	1	5.0
ING-IND/22	PHYSICAL PROPERTIES OF MOLECULAR MATERIALS	1	5.0
ING-INF/01	QUANTUM CIRCUITS AND DEVICES	1	5.0
ING-INF/01	SENSOR SYSTEMS	1	5.0
ING-INF/04	SYSTEMS THEORY (NONLINEAR DYNAMICS)	1	5.0
ING-IND/22	ADVANCED MATERIALS	1	10.0
ING-INF/04	ADVANCED AND MULTIVARIABLE CONTROL	2	10.0
ING-INF/01	BIOCHIP	2	5.0
ING-INF/01	ELECTRONICS AND ELECTROACOUSTICS FOR SOUND ENGINEERING - MODULE 2: ELECTROACOUSTICS FOR SOUND ENGINEERING	2	5.0
ING-INF/01	ELECTRONICS DESIGN FOR BIOMEDICAL INSTRUMENTATION	2	10.0

ING-IND/18	FISSION REACTOR PHYSICS	2	10.0
ING-INF/04	FONDAMENTI DI ROBOTICA	2	5.0
ING-IND/18	HIGH INTENSITY LASERS FOR NUCLEAR AND PHYSICAL APPLICATIONS II	2	5.0
ING-INF/05	MACHINE LEARNING	2	5.0
ING-INF/01	MICROCONTROLLORI	2	5.0
ING-INF/01	MICROELECTRONIC TECHNOLOGIES	2	5.0
ING-IND/22	MOLECULAR MODELING OF MATERIALS	2	5.0
ING-INF/04	NATURAL RESOURCES MANAGEMENT	2	10.0
ING-IND/18	NUCLEAR TECHNIQUES FOR THE ANALYSIS OF MATERIALS	2	5.0
ING-INF/02	PHOTONIC COMPUTING	2	5.0
ING-INF/01	RADIATION DETECTION SYSTEMS	2	5.0
ING-IND/22	SCIENCE AND ENGINEERING FOR THE CONSERVATION OF CULTURAL HERITAGE	2	5.0

Groups INT and MAN

INT (Interdisciplinary)

SSD	Denominazione Insegnamento	Lingua	Tipo	Sem	CFU
MED/50	METHODS AND TECHNOLOGIES FOR NEUROSENSORY SYSTEMS *Studenti da A a ZZZZ - Barbieri Riccardo*	EN	M	1	6.0
ICAR/13	PROFESSIONAL WORKSHOP SWT [(a)] *Studenti da A a ZZZZ - Docente non definito*	EN	M	1 ([2])	6.0
ING-INF/06	LABORATORY OF WEARABLE BIOSIGNAL PROCESSING [(b)] *Studenti da A a ZZZZ - Docente non definito*	EN	M	2	5.0
ING-IND/34	SMART MATERIALS FOR WEARABLES *Studenti da A a ZZZZ - Fare' Silvia*	EN	M	2	5.0
ING-INF/05	WEARABLE SYSTEMS LAB [(c)] *Studenti da -- a -- - Docente non definito*	EN	M	2	5.0 [1.0 di]
ING-INF/03	WIRELESS SENSOR BODY AREA NETWORKS *Studenti da A a ZZZZ - Magarini Maurizio*	EN	M	2	5.0 [1.0 di]

nel secondo emisemestre

MAN (Management)

SSD	Denominazione Insegnamento	Sem	CFU
ING-IND/35	ECONOMIA E ORGANIZZAZIONE AZIENDALE C	2	5.0
ING-IND/35	HIGH-TECH ENTREPRENEURSHIP	2	5.0
ING-IND/35	MANAGEMENT OF INNOVATION PROJECTS	2	10.0
ING-IND/35	PATENTS AND INTELLECTUAL PROPERTY MANAGEMENT	2	5.0

Group M&D (Materials and Devices)

SSD	Denominazione Insegnamento	Sem	CFU
ING-IND/22	CORROSION ENGINEERING	1	5.0
ING-IND/22	FUNCTIONAL MATERIALS	1	5.0
ING-IND/22	FUNDAMENTALS OF MATERIALS SCIENCE	1	5.0
ING-INF/01	MEMS SYSTEM AND ELECTRONICS - PART I (INERTIAL)	1	5.0
ING-INF/01	MEMS SYSTEM AND ELECTRONICS - PART II (MAGNETIC, ACOUSTIC, IMAGING)	1	5.0
ING-IND/13	MICRO AND NANO ROBOTICS	1	5.0
FIS/01	NANOELECTRONICS OF GRAPHENE AND RELATED 2D MATERIALS	1	5.0
ING-IND/22	NANOSTRUCTURED MATERIALS	1	5.0
FIS/01	SEMICONDUCTOR QUBITS	1	5.0
ING-INF/01	SENSOR SYSTEMS	1	5.0
ING-IND/21	SURFACE TREATMENTS FOR ENGINEERING APPLICATIONS	1	5.0
ING-IND/22	VISCOELASTIC BEHAVIOUR OF POLYMERS	1	5.0
ING-INF/01	BIOCHIP	2	5.0
FIS/01	FERROIC MATERIALS	2	5.0
ING-IND/23	MATERIALS FOR ELECTRONICS	2	5.0
ICAR/08 ING-IND/34	MECHANICAL TESTING AT SMALL SCALE	2	5.0
ICAR/08	MECHANICS OF MATERIALS AND INELASTIC CONSTITUTIVE LAWS	2	5.0
ICAR/08	MEMS MECHANICS AND DESIGN – ACTUATORS	2	5.0
ICAR/08	MEMS MECHANICS AND DESIGN - SENSORS	2	5.0
ING-IND/34	MICROMECHANICS	2	5.0
ING-IND/21	SMART MATERIALS	2	5.0

Groups MATH and MOD

MAT (Mathematics)

SSD	Denominazione Insegnamento	Sem	CFU
MAT/05	CALCULUS OF VARIATIONS	1	5.0
MAT/07	MATHEMATICS OF QUANTUM MECHANICS	1	10.0
MAT/05	REAL AND FUNCTIONAL ANALYSIS	1	5.0
MAT/06	STOCHASTIC DIFFERENTIAL EQUATIONS	1	5.0
MAT/06	STOCHASTIC DYNAMICAL MODELS	1	5.0
MAT/03	TOPOLOGIA ALGEBRICA COMPUTAZIONALE	1	8.0
MAT/05	ADVANCED PARTIAL DIFFERENTIAL EQUATIONS	2	5.0
SECS-S/01	APPLIED STATISTICS	2	5.0
MAT/03	GEOMETRIA DIFFERENZIALE	2	8.0
MAT/02	GROUP THEORY	2	5.0
MAT/07	METHODS AND MODELS FOR STATISTICAL MECHANICS	2	5.0
MAT/05	OPERATOR THEORY	2	5.0

MOD (Modeling)

SSD	Denominazione Insegnamento	Sem	CFU
ING-INF/03	ADVANCED DIGITAL SIGNAL PROCESSING A	1	5.0
ING-INF/03	ADVANCED DIGITAL SIGNAL PROCESSING B	1	5.0
ICAR/08	COMPUTATIONAL MODELING FOR MATERIALS ENGINEERING	1	5.0
MAT/08 SECS-S/01	COMPUTATIONAL STATISTICS	1	5.0
ICAR/08	MULTIPHYSICS MODELLING AND SIMULATION	1	5.0
MAT/08	ADVANCED PROGRAMMING FOR SCIENTIFIC COMPUTING A	2	5.0
MAT/08	ADVANCED PROGRAMMING FOR SCIENTIFIC COMPUTING B	2	5.0

Groups PHYS-1,2 (Physics)

PHYS-1

SSD	Denominazione Insegnamento	Sem	CFU
FIS/01	ADVANCED MICRO AND NANOFABRICATION TECHNOLOGIES	1 (2)	5.0
FIS/01	ANALOG AND DIGITAL HOLOGRAPHY	1	5.0
FIS/01	APPLICATIONS OF ULTRAFAST PROCESSES	1	5.0
FIS/01	BIOMEDICAL OPTICS	1	5.0
FIS/01	ELECTRON AND SCANNING PROBE MICROSCOPY	1 (1)	5.0
FIS/01	MAGNETISM AND SUPERCONDUCTIVITY	1	5.0
FIS/01	NANOELECTRONICS OF GRAPHENE AND RELATED 2D MATERIALS	1	5.0
FIS/01	NANOMAGNETISM AND SPINTRONICS	1	5.0
FIS/01	OPTICAL MICROSCOPY	1	5.0
FIS/01	PHYSICS AND TECHNOLOGY OF OPTICAL MATERIALS	1	5.0
FIS/01	PHYSICS OF ANTIMATTER AND APPLICATIONS	1	5.0
FIS/01	PHYSICS OF ULTRAFAST PROCESSES	1	5.0
FIS/01	SEMICONDUCTOR NANOSTRUCTURES	1	5.0
FIS/01	SEMICONDUCTOR QUBITS	1	5.0
FIS/01	BIOPHOTONICS (C.I.)	1	10.0
FIS/01	IMAGING AND FABRICATION OF MICRO AND NANOSTRUCTURES (C.I.)	1	10.0
FIS/01	MAGNETISM, SUPERCONDUCTIVITY AND SPINTRONICS [I.C.]	1	10.0
FIS/01	PHYSICS AND APPLICATIONS OF ULTRAFAST PROCESSES (C.I.)	1	10.0
FIS/01	QUANTUM DEVICES AND NANOELECTRONICS -	1	10.0
FIS/01	ADVANCED TOPICS IN ENGINEERING PHYSICS	2	5.0

PHYS-1

FIS/01	ATOM OPTICS AND QUANTUM TECHNOLOGIES	2	5.0
FIS/01	COMPLEMENTI DI MECCANICA QUANTISTICA	2	5.0
FIS/01	COMPUTER-ASSISTED OPTICAL SYSTEM DESIGN	2	5.0
FIS/01	FERROIC MATERIALS	2	5.0
FIS/01	INTEGRATED PHOTONICS	2	5.0
FIS/01	META PHOTONICS	2	5.0
FIS/01	PHYSICS OF PHOTOVOLTAIC PROCESSES	2	5.0
FIS/01	QUANTUM OPTICS AND INFORMATION	2	5.0
FIS/01	QUANTUM PHENOMENA IN LOW DIMENSIONAL SYSTEMS	2	5.0
FIS/01	RELATIVITY	2	5.0

FIS/01	STATISTICAL PHYSICS OF COMPLEX SYSTEMS	2	5.0
FIS/01	SYMMETRY APPLICATIONS IN CLASSICAL AND QUANTUM PHYSICS	2	5.0
FIS/01	INTEGRATED AND META PHOTONICS [I.C.] -	2	10.0
FIS/01	QUANTUM OPTICS AND QUANTUM TECHNOLOGIES (C.I.) -	2	10.0

PHYS-2

SSD	Denominazione Insegnamento	Sem	CFU
FIS/03	PHYSICS OF DISORDERED MATERIALS	1	5.0
FIS/03	PLASMA PHYSICS I+II	1	10.0
FIS/03	STATISTICAL PHYSICS	1	10.0
FIS/01 ING-IND/34	APPLICATIONS OF OPTICAL MICROSCOPY IN BIOMEDICINE	2	5.0
FIS/01	METHODS FOR MULTIMODAL FUNCTIONAL EVALUATION OF THE CENTRAL NERVOUS SYSTEM	2	5.0

Groups SOFT

SSD	Denominazione Insegnamento	Sem	CFU
FIS/01	INTERNATIONAL PHYSICS COMPETITIONS	1	5.0
L-FIL- LET/11	ITALIAN AND EUROPEAN CULTURE	1	5.0 [5.0 di]
M-FIL/02	PHILOSOPHY OF SCIENCE AND TECHNOLOGY	1	5.0 [5.0 di]
M-FIL/02	AI AND SOCIETY	2	5.0
M-FIL/02	BIOMEDICAL TECHNOLOGIES: PHILOSOPHICAL AND ETHICAL ISSUES	2	5.0 [5.0 di]
M-PED/03	COMMUNICATION AND ARGUMENTATION	2	5.0 [5.0 di]
M-FIL/02	CRITICAL THINKING	2	5.0 [5.0 di]
M-FIL/02	DATA GOVERNANCE AND ETHICS	2	5.0
SPS/07	EMERGING TECHNOLOGIES AND SOCIETAL CHALLENGES	2	5.0 [5.0 di]
ING-INF/07	ENGINEERING METHODS SUPPORTING JUSTICE	2	5.0 [5.0 di]
M-FIL/02	ETHICS FOR TECHNOLOGY	2	5.0 [5.0 di]
M-STO/04	HISTORY OF TECHNOLOGICAL INNOVATION	2	5.0 [5.0 di]
FIS/01	INTERNATIONAL PHYSICS COMPETITIONS	2	5.0
ING-INF/05	PERSONALITÀ, TEAM BUILDING, LEADERSHIP	2	5.0 [5.0 di]
SPS/08	SOCIAL HISTORY OF TECHNOLOGY AND DIGITAL INFRASTRUCTURES	2	5.0 [5.0 di]
ING-IND/10	SUSTAINABLE DEVELOPMENT	2	5.0 [5.0 di]

SPS/08	THE SOCIAL SHAPING OF TECHNOLOGY	2	5.0 [5.0 di]
ING-INF/04 SPS/09	DIVERSITY AWARE DESIGN OF TECHNOLOGY SOLUTIONS -	2	5.0 [5.0 di]

Management of Innovation Projects

Prof Raffaella Cagliano

Interdisciplinary Programme in Smart Wearable Technologies (SWT)

SMART EYEWEAR LAB | EssilorLuxottica | POLITECNICO MILANO 1863

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The Interdisciplinary Programme in SWT is a set of courses (including several laboratory courses) that characterize the Master's Degree path, aiming to enhance skills for entry into the smart wearables industry.

It provides in-depth multidisciplinary experience, specialized knowledge in smart wearables, and the ability to collaborate effectively with professionals from various sectors.

It is available for all the four tracks of *Engineering Physics*.

Students participating in the program will be able to dedicate at least 30 ECTS to specialized training in the field of smart wearables, within the 120 ECTS required for their degree program (with no extra ECTS).

Interdisciplinary Programme in Smart Wearable Technologies (SWT)

SMART EYEWEAR LAB | EssilorLuxottica | POLITECNICO MILANO 1863

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Table A - 15 Credits

Applied Physics

SSD	*Course Title*		Semester	Credits
FIS/01	**ANALOG AND DIGITAL HOLOGRAPHY**	*NEW*	1	5
FIS/01	**PHYSICS AND TECHNOLOGY OF OPTICAL MATERIALS**	*NEW*	1	5
FIS/01	ADVANCED MICRO AND NANOFABRICATION TECHNOLOGIES		1	5
FIS/01	COMPUTER-ASSISTED OPTICAL SYSTEM DESIGN		2	5
FIS/01	META PHOTONICS		2	5

Table B - 15 Credits

Electronics

Computer Science

Biomedical Engineering

Design

SSD	*Course Title*		Semester	Credits
ING-INF/01	**SMART WERABLE DESIGN AND PROTOTYPING**	*NEW*	2	5
ING-INF/01	SENSOR SYSTEMS		1	5
ING-INF/01	BIOCHIP		2	5
ING-INF/05	**SOFTWARE ARCHITECTURES FOR WEARABLE AND PERVASIVE SYSTEMS**	*NEW*	2	5
ING-INF/05	MACHINE LEARNING		2	5
ING-INF/05	DIGITAL SYSTEMS DESIGN		1	10
ING-INF/05	ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING		1	5
ING-INF/04	MODEL IDENTIFICATION AND DATA ANALYSIS		1	10
ING-INF/05	STREAMING DATA ANALYTICS		1	5
ING-INF/05	**WEARABLE SYSTEMS LAB**	*NEW*	2	5
ING-INF/03	**WIRELESS SENSOR BODY AREA NETWORKS**	*NEW*	2	5
ING-IND/34	**SMART MATERIALS FOR WEARABLE**	*NEW*	2	5
ING-INF/06	**LABORATORY OF WEARABLE SIGNAL PROCESSING**	*NEW*	2	5
MED/50	METHODS AND TECHNOLOGIES FOR NEUROSENSORY SYSTEMS		1	6
ICAR/13	**PROFESSIONAL WORKSHOP**	*NEW*	1	6

Admissions

Coming from Ingegneria Fisica of POLIMI there is only one rule:

Average grade >= 21.5

This threshold is a rigid requirement, exceptions are **very very very unlikely**.

Contact jointly Prof. Ghiringhelli, Dallera and Zavelani Rossi to enquire about possible exceptions if you believe you are eligible for exceptional reasons

Study Plans

If you do not like suggested traks/PSPA, you can submit an **Individual Study Plan** (***Piano degli Studi Autonomo***)

- Prof. Dallera and Zavelani Rossi will evaluate it.
- It might be approved or not.
- Mandatory courses cannot be eliminated or replaced
- Quotas of Physics, Mathematics, Engineering must be respected
- In all cases minimum requirements from the law and the POLIMI rules must be fulfilled (next slides).
- Before submitting the Individual Study Plan send an e-mail to both Dallera and Zavelani or a pre-evaluation; explain clearly the motivations of your choice.
- Use only your POLIMI e-mail account for all communication with professors and offices.

Study Plans

FAQs

- No changes about 1st semester can be made in February
- Exams in “sovrannumero” “S”(extra exams): at your choice, possible changes (with some rules)
- Extra exams up to 75 CFU/year: no extra taxes
- Courses with restricted number of students (Numero chiuso): fist-come-first-serve (you find them only in non-mandatory courses)
- Interneships (tirocini): possible but in addition and not in substitution of exams

LM-44 Classe delle lauree magistrali in MODELLISTICA MATEMATICO-FISICA PER L'INGEGNERIA

ATTIVITÀ FORMATIVE INDISPENSABILI
Attività formative caratterizzanti

Discipline matematiche, fisiche e informatiche
Metodi avanzati della matematica, dell'informatica e della fisica classica e dei fondamenti della fisica moderna; modelli matematici, metodi di simulazione e di calcolo numerico e simbolico.
FIS/01 - Fisica sperimentale FIS/02 - Fisica teorica, modelli e metodi matematici
FIS/03 - Fisica della materia FIS/04 - Fisica nucleare e subnucleare
INF/01 - Informatica MAT/02 - Algebra MAT/03 - Geometria
MAT/05 - Analisi matematica MAT/06 - Probabilità e statistica matematica
MAT/07 - Fisica matematica MAT/08 - Analisi numerica MAT/09 - Ricerca operativa

Discipline ingegneristiche
Modelli matematici e tecniche di simulazione e di calcolo in almeno un'area dell'ingegneria. ICAR/01 - Idraulica
ICAR/08 - Scienza delle costruzioni ING-IND/06 - Fluidodinamica
ING-IND/10 - Fisica tecnica industriale
ING-IND/13 - Meccanica applicata alle macchine
ING-IND/18 - Fisica dei reattori nucleari
ING-IND/22 - Scienza e tecnologia dei materiali
ING-IND/31 - Elettrotecnica ING-INF/01 - Elettronica
ING-INF/02 - Campi elettromagnetici ING-INF/04 - Automatica
ING-INF/05 - Sistemi di elaborazione delle informazioni

***Numero minimo di CFU riservati alle attività caratterizzanti* 45**

Vincoli di Statuto (da Regolamento)

The Master of Science in Engineering Physics belongs to the **LM-44 class** – *Mathematical-Physical Modeling for Engineering*.

This degree class requires at least **45 ECTS credits** in **Core Educational Activities**, of which:

- **At least 30 ECTS** must be in **mathematics, physics, and computer science disciplines**,
- **At least 15 ECTS** must be in **engineering**.

ambito disciplinare	settore	CFU		minimo da D.M. per l'ambito
		min	max	
Discipline matematiche, fisiche e informatiche	FIS/01 Fisica sperimentale FIS/03 Fisica della materia MAT/05 Analisi matematica MAT/06 Probabilità e statistica matematica MAT/07 Fisica matematica MAT/08 Analisi numerica	30	50	
Discipline ingegneristiche	ING-IND/18 Fisica dei reattori nucleari ING-IND/22 Scienza e tecnologia dei materiali ING-INF/01 Elettronica ING-INF/02 - Campi elettromagnetici ING-INF/04 Automatica ING-INF/05 Sistemi di elaborazione delle informazioni	15	40	
Minimo di crediti riservati dall'ateneo minimo da D.M. 45:		-	45 - 90	

International mobility: procedures

- **November 2025**: first application for international mobility
- **April 2026**: second application for international mobility
- Applications via **online services** of Politecnico di Milano
- Choose an institution with **academic courses compatible** with the tracks of Engineering Physics
- Once selected, **FIRST** prepare a POLIMI study plan to be submitted to prof. Zavelani-Rossi and prof. Dallera for approval,
  THEN prepare the online learning agreement (OLA).
- CONTACT for OLA approval:
  - prof. Bottegoni for ***Quantum Materials and Nanophysics*** track
  - prof. Lucchini for ***Photonic and Quantum Technologies*** track
  - prof. Bottegoni or prof. Lucchini for ***Micro- and Nano-Systems*** **and** ***Mathematics and Physics for Quantum Engineering*** **tracks**
  - prof. Bassi for ***double degree***